

Neurochemistry and post-mortem neurotransmitters: Toward the study of neurochemical connectivity

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ABSTRACT

Background: Monoaminergic systems, as some other neuromodulators, exert modulatory influences across the entire central nervous system (CNS), participating in basic neurobiological processes, behavioural traits, neuropsychiatric diseases, and responses to psychotropic drugs. Their activity is supposedly balanced across the CNS i.e. a change of input in one region could alter their activities at distal regions. Unfortunately, there are a few attempts to determine the neurochemical relationships between local or distal regions.

New method: In this review article, we present the technical requirements for performing complex analysis of neurochemical, postmortem, quantitative data on neurotransmitter systems. We explore new types of neurochemical analysis beyond quantitative differences to approximate neurochemical connectivity.

Results: Although the lack of temporality and unclear meaning of neurotransmitter tissue content are limitations of this approach, postmortem measurement offers the possibility of quantitatively and qualitatively assessing the content of monoamines and/or metabolites in multiple parts of the CNS. This form of neurochemical connectivity is distinct from anatomical connection and brain functional connectivity.

Comparison with existing methods: This expands the study of neurotransmitter systems beyond a specific/limited set of brain regions and allows for the description of possible inter-regional influences within a neurotransmitter system and between neurotransmitter systems across the CNS.

Conclusion: Through this unprecedented discussion, this article advances the use of neurochemical data towards multiple dimensions of analysis, primarily based on monoaminergic systems and their neurochemical connectivity.

1. Introduction

Monoaminergic systems, including dopamine (DA), serotonin (5-HT), and noradrenaline (NA), play a key role in virtually all cerebral functions. Their extensive innervation across the central nervous system (CNS), especially by 5-HT and NA neurons, implies multiple actions in partnership with other neurotransmitter systems. Modification of their activity has been identified in most, if not all, neuropsychiatric and neurological diseases (Di Giovanni et al., 2016), such as in depression, anxiety, schizophrenia, or substance abuse, although the nature of these

monoaminergic modifications are not clear. The benefit derived from monoaminergic-based treatments such as in depression, anxiety, schizophrenia, or Parkinson's disease, suggests that defects of monoaminergic systems are causally involved in these pathologies. This is an oversimplified view of monoaminergic systems, as monoaminergic disturbances in other diseases, including Alzheimer's disease, addiction, and epilepsy, have not yet resulted in approved monoamine-based treatments (De Deurwaerdère and Di Giovanni, 2017; De Deurwaerdère and Di Giovanni, 2020; Di Giovanni et al., 2016; Simic et al., 2017; Svob Strac et al., 2016; Tomkins and Sellers, 2001). The

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reality is that we still do not entirely understand the function of monoaminergic systems.

There are several ways to study the activity of monoaminergic systems including electrophysiology, anatomy, behaviour, and biochemistry. Post-mortem measurements of monoamines and their metabolites have been used for several years to approximate the function of monoaminergic neurons. These measures have provided a means to determine the functional dynamics of monoaminergic systems in various physiological conditions, environmental and pharmacological contexts, pathologies, drug mechanisms, and animal models of brain diseases. This kind of measurement can be performed in vertebrates and invertebrates, as the function of monoaminergic systems (mostly DA and 5-HT) is largely conserved across the animal kingdom (Bacque-Cazenave et al., 2020; Cenci and Björklund, 2020; Jacobs and Azmitia, 1992; Smidt and Burbach, 2009). There are differences in monoamine metabolism across species, and even strong regional differences within a given species. However, the general idea of their functioning remains the same: volume transmission, a balance between gland and neuron-like function (Descarries and Mechawar, 2000). These tissue parameters are used to infer neuronal electrical activity or neurotransmitter release (Aby et al., 2022; Shannon et al., 1986). Nonetheless, the links between the electrical activity, neurotransmitter release, and tissue quantity are still enigmatic, and interpretation between these parameters is always speculative. Currently, post-mortem tissue measurements are not widely used as they only give a static measure of monoaminergic parameters, in contrast to electrophysiology, push-pull cannula, intracerebral microdialysis, and more recently, biosensor techniques. Conversely, the aforementioned *in vivo* approaches cannot offer a widespread picture of monoaminergic function in the CNS due to their invasive nature, and the inherent reduction of sampled brain regions. Considering that regional changes in monoamine tissue markers are frequently observed in animal models under specific conditions, the balance of monoamines within and between regions appears to be a critical characteristic of monoaminergic systems (Bharatiya et al., 2020; Fitoussi et al., 2013; Mlost et al., 2024). Being quantitative and offering the possibility to span multiple regions, post-mortem parameters are interesting markers for exploring regional links within and between neurotransmitter systems across the CNS using correlative and multivariate analyses. These links would correspond neither to anatomical connectivity nor functional connectivity, and will be thus termed neurochemical connectivity.

In this review, we explore the diverse conditions under which tissue monoamines and related compounds are measured, with a focus on the analysis and interpretation of these indices. Taking into account the widespread innervation, morphology and regulation of monoaminergic systems, and capitalizing on 15 years of in-house data, we broaden classical region-specific quantitative analysis, to evaluate monoamine tissue content across the brain, leveraging multiple correlations within and between monoaminergic systems and other neurochemical systems. We believe this is one way to approximate neurochemical connectivity within and between neurotransmitter systems - before conceiving other types of analyses including multivariate analysis and deep learning, discussed elsewhere (Parrot, 2025). These studies are still a complementary analysis of the quantitative analysis before being considered as a true hypothesis-generating framework.

2. Gross anatomy of monoaminergic neurons

Monoaminergic neurons project from brainstem nuclei throughout the brain and spinal cord. These projections include DA neurons located in the ventral tegmental area (VTA) and the substantia nigra pars compacta (SNpc), NA neurons arising at least from locus coeruleus (LC), and 5-HT neurons originating from the median and dorsal raphe nuclei (MRN and DRN) (Aston-Jones, 2004; Björklund and Dunnett, 2007; Gerfen et al., 1987; Hale and Lowry, 2011; Steinbusch, 1984). Yet, the monoaminergic innervation of the CNS is more complex, and it conditions their quantities and metabolism.

2.1. Dopaminergic system

DA neurons are a heterogeneous group of cells localized in the mesencephalic, diencephalic and olfactory bulb regions of the brain. However, most DA cells reside in the ventral part of the mesencephalon (Björklund and Dunnett, 2007). The major DA pathways are the nigrostriatal and the mesolimbic that originate from mesodiencephalic DA neurons corresponding to A8, A9 and A10 cell groups. The nigrostriatal pathway corresponds to DA neurons originating from the lateral SNpc (A9 cell group) that project into the striatum (caudate nucleus and putamen). These neurons are widely studied because their degeneration is involved in the development of Parkinson's disease (Cenci and Björklund, 2020; Grace and Bunney, 1985). The VTA DA cell bodies (A10 cell group) send projections to the nucleus accumbens (NAc), the ventromedial striatum, and extend towards structures closely associated with the limbic system (Deutch et al., 1987; Di Giovanni et al., 2009; Gerfen et al., 1987; Kalivas, 1993; White, 1996). The mesocortical DA pathway, also originating from VTA, innervates cortical regions (Bannon and Roth, 1983; Deutch, 1993; Tassin et al., 1976).

The substantia nigra (SN) also comprises the pars reticulata (SNpr) area which mainly contains γ -amino-butyric acid (GABA)-ergic neurons, constituting the major efferent projections of the basal ganglia. Nonetheless, the SNpr also contains some DA neurons that project to the striatum (Björklund and Dunnett, 2007; Matsuda et al., 2009). There are also less defined DA neurons in the A11 nucleus that are responsible for a sparse DA innervation of the spinal cord (Barraud et al., 2010; Koblinger et al., 2014). Other DA centers are found in the hypothalamus (A12 nucleus), playing a role in neuroendocrine processes via projection to the pituitary gland (tuberoinfundibular pathway) (Björklund and Dunnett, 2007). Some DA neurons have been reported in the DRN (Halberstadt and Balaban, 2008; Stratford and Wirtshafter, 1990).

2.2. Serotonergic system

5-HT innervation in the CNS is widespread, with most CNS territories receiving projections to varying degrees. This widespread distribution originates from 5-HT cell bodies in the MRN and DRN (Azmitia and Segal, 1978; Steinbusch, 1984). The DRN is the brain region housing the highest proportion of 5-HT neurons, corresponding to 30–56% of 5-HT neurons in the CNS, depending on the species (Jacobs and Azmitia, 1992). 5-HT neurons may co-express and release other neurotransmitters, such as GABA, glutamate, and nitric oxide, in addition to neuropeptides, such as corticotropin-releasing factor, substance P, or cholecystokinin (De Deurwaerdere and Di Giovanni, 2017; Jacobs and Azmitia, 1992; Ren et al., 2019). Regarding glutamate, it has been well described that 5-HT neurons express the glutamate vesicular transporter Vglut3 (Amilhon et al., 2010). The 5-HT-containing neurons of the DRN send projections to DA neurons in both the VTA and SNpc, with their terminal fields including the NAc, mPFC, amygdala, and striatum (De Deurwaerdere and Di Giovanni, 2017). They also innervate midbrain regions housing NA neurons, including the locus coeruleus (LC) (Aston-Jones et al., 1991; Muzerelle et al., 2016; Steinbusch, 1984).

DRN 5-HT neurons are heterogeneous and variably express proteins such as vGlut3 (Commons, 2020; Okaty et al., 2020, 2015; Ren et al., 2019). These data are consistent with previous anatomical studies describing at least six distinct pathways from the DRN (Hale and Lowry, 2011): the dorsal part projects to the central and basolateral nuclei of the amygdala, the dorsal hypothalamic area, and the mPFC. The ventral DRN projects to the sensorimotor cortex and caudate putamen; the lateral DRN projects to the lateral hypothalamus or superior colliculus; the rostral DRN projects to the caudate putamen and SN; and the caudal DRN projects to the amygdala, ventral hippocampus, and thalamic nuclei (Lowry et al., 2008; Steinbusch, 1984).

The MRN contains fewer 5-HT cell bodies than the DRN, approximately 5% of which project to the hippocampus (Ren et al., 2019; Steinbusch, 1984, 1981). Although the MRN innervates several brain

regions, the projections from the MRN to at least the basal ganglia or mPFC presumably do not release 5-HT (De Deurwaerdere and Di Giovanni, 2017; Jacobs and Azmitia, 1992). MRN 5-HT projections are primarily responsible for 5-HT innervation in brain regions such as the dorsal hippocampus and lateral septum. Other 5-HT cell bodies from the nucleus reticularis tegmenti pontis, raphe pontis, and raphe magnus can innervate the medial septum (Köhler et al., 1982). 5-HT projections could also originate from the raphe obscurus (Vertes and Martin, 1988), although the 5-HT-ergic nature of these neurons has not been clearly established, as suggested in opossums (Martin et al., 1985).

Considering this distribution, the measurement of 5-HT and its metabolites in a piece of tissue corresponds to the overlap of fibres originating from the DRN and MRN in several brain regions.

2.3. Noradrenergic system

NA cell bodies are distributed across 7 distinct nuclei in the brainstem (A1–7) and innervate almost all brain regions. The NA system primarily corresponds to LC (A6). A1, A5, and A7 clusters are responsible for some ascending and ventral NA projections to the septum, bed nucleus of the stria terminalis, and hypothalamus (HY). The LC, which contains approximately 50% of adrenergic neurons, is present in all mammalian species and is located in the dorsal pons. It is mainly responsible for adrenergic innervation of the telencephalon and represents the main source of NA within the CNS (Aston-Jones, 2004). The widespread network of LC-derived adrenergic projections exerts a powerful modulatory influence on both cortical and subcortical circuits (Chandler et al., 2019). Conversely, adrenergic innervation of the striatum and globus pallidus is sparse (Aston-Jones, 2004), consistent with lower tissue NA levels in these regions (Fitoussi et al., 2013).

2.4. Morphology of monoaminergic neurons

Beyond the number of cell bodies located in each monoaminergic centre, there is the issue of axonal branches and the shape of the fibres. It is now well demonstrated that monoaminergic fibres have an extremely broad axonal development, with numerous “en passant” synapses that serve as potential sites of uptake and release (Descarries et al., 1991; Descarries and Mechawar, 2000). Matsuda et al. (2009) successfully labelled one dopaminergic neuron from the SN pars reticulata projecting to the striatum (Matsuda et al., 2009). It is remarkable that a single neuron can project to a large portion of the striatum, spanning functionally distinct territories, such as the matrix and striosome. The reconstruction of individual DRN 5-HT neurons has demonstrated more extensive axonal projections throughout the brain than previously thought (Ren et al., 2019, 2018).

During the late 1960's and 1970's, neurochemists identified distal pools of neurotransmitter release located away from traditional sites using pharmacological compounds (Glowinski, 1975). The anatomical counterpart to these neurochemical data came years later with evidence that certain varicosities, despite housing vesicles of exocytosis, cannot release their contents (Liu et al., 2018). This has only been described for DA axons in the striatum, which could explain the relatively low DA metabolism in the striatum compared to other brain regions (see below). Whether the findings of Liu et al. (2018) apply to DA neurons in extrastriatal territories and 5-HT and NA neurons remains unknown.

2.5. Influence of monoaminergic systems on executive neurotransmission

Monoaminergic systems are strongly interconnected at various levels, from the soma to the distal projections across the brain (De Deurwaerdere and Di Giovanni, 2017). Although most quantitative studies show the effects of drugs on monoaminergic systems in various brain regions, they do not explain how these monoaminergic modifications influence each other or the regional and global executive systems that compose the functional networks using glutamate, aspartate,

and GABA as neurotransmitters. These interactions, both inter-neurotransmitter and inter-regional, are labile and are influenced by specific stimuli and tasks engaged (Butler et al., 2025a,b).

Another level of complexity could be attributed to their modulatory action on executive systems (Fig. 1). The influence of monoamines on these systems occurs at different anatomical levels within and between functional networks. Conversely, these executive networks could directly interact with the monoaminergic systems in targeted regions or acting on neurotransmitter release via feedback to the monoaminergic nuclei. This distributed modulatory action has been largely overlooked, even though it looks to be the main functional property of monoaminergic neurons.

Monoamines are rarely used to elaborate anatomo-functional models of brain operations. An exception is the case of DA neurons from the midbrain and their receptors that entered the anatomo-functional model of basal ganglia (Albin et al., 1989; DeLong, 1990). This is in part due to the distribution of DA receptors on the direct (D1 receptors) and the indirect pathways (D2 receptors), and the pathophysiology of the basal ganglia and L-DOPA-induced dyskinesia. This model, that is still taught, was strongly discussed soon after its publication, as there are several reasons to take distance from the model. For instance, the segregation of DA receptors was not clearly evident (Gerfen and Surmeier, 2011), and the pathophysiological aspects of these models were based on the assumption that L-DOPA-induced dyskinesia was associated with excessive DA release in the striatum, although the opposite may be true (De Deurwaerdere et al., 2017; De La Fuente-Fernandez, 2004; Porras et al., 2014). Meanwhile, as far as DA could be integrated in the models, the striatal part was incorporated into the integrated actions of DA in the whole basal ganglia and the cortex (Obeso et al., 2000). This approach remains unsatisfactory, but highlights the difficulty in integrating neuromodulatory systems into defined frameworks.

The use of other techniques is essential for better understanding this matrix of interactions and its dysregulation. Post-mortem measurements of monoamines across a plethora of regions, using validated and calibrated techniques, offer an approach to overcome this challenge.

3. Biochemistry of monoaminergic systems

3.1. Biochemical organization of monoaminergic neurons

Like other systems of neurotransmission, the biochemical organization of monoaminergic neurons remains complex, despite receiving considerable attention. The first layer of complexity arises from the conception of neurons as polarized cells, from the somatodendritic regions to axons. But when addressing the biochemical function of the neurons, this polarization vanishes towards compartments capable of both synthesizing and releasing the neurotransmitter to varying degrees (Ludwig et al., 2016). Somatodendritic release of monoamines was shown *in vivo* in 1981 for DA (Cheramy et al., 1981), and later for all monoaminergic neurons (Adell et al., 1993; Adell and Artigas, 1998; Hery and Ternaux, 1981; Pudovkina and Westerink, 2005; Quentin et al., 2018; Westerink et al., 1987). This represents a fundamental mechanism participating in the regulation of neuronal activity and neurotransmission.

The second layer of complexity arises from the multiple enzymes and transporters maintaining the biochemical activity of neuronal compartments in a steady state of non-equilibrium, which could correspond to homeostasis (Justice, 1993). Fig. 2 reports the general organization of the anabolic and catabolic pathways of monoaminergic neurons. The metabolism involves monoaminergic and non-monoaminergic neurons, and glial cells. The catecholamines - DA and NA share the initial involvement of cytosolic tyrosine hydroxylase (TH) in the biosynthesis of endogenous L-DOPA. L-DOPA is then decarboxylated into DA by aromatic amino acid decarboxylase (AADC) in the cytosol of catecholaminergic neurons. DA enters the vesicles of exocytosis via the vesicular monoamine transporter type 2 (VMAT2), facilitated by the acidic

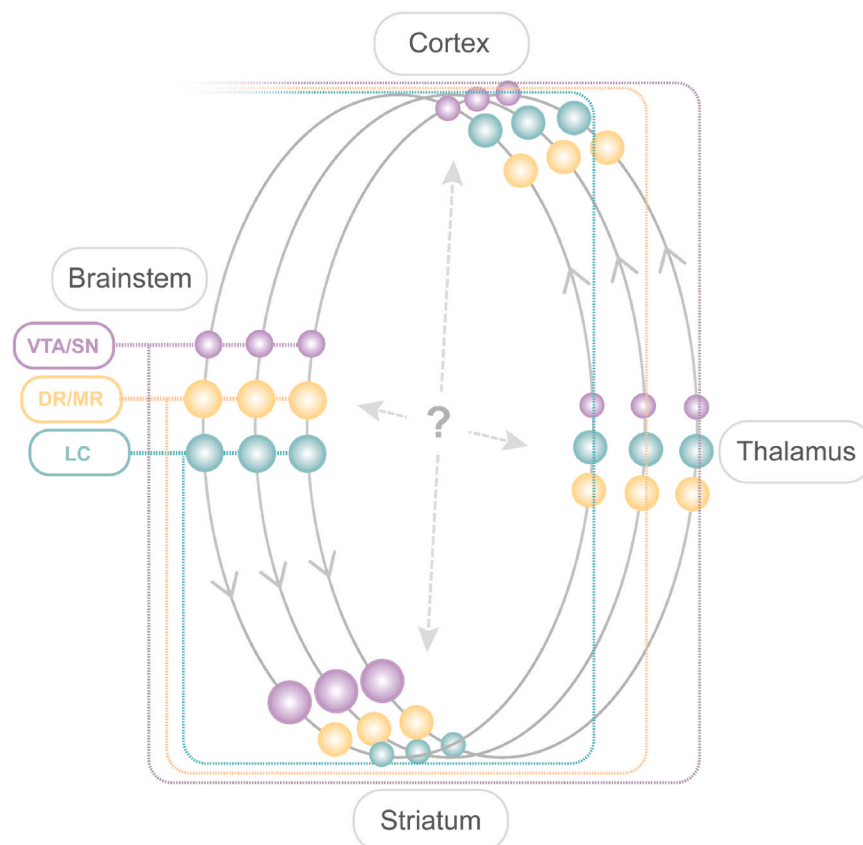


Fig. 1. Interaction between monoamines and executive neurotransmitter systems. Monoaminergic neurons originate in the brainstem and innervate in a diffuse manner numerous interconnected brain regions. These axonal projections are depicted as dashed lines from the dorsal and median raphe nucleus (DR and MR; 5-HT axons yellow); locus coeruleus (LC; NA axons teal); substantia nigra and ventral tegmental area (SN and VTA; DA axons purple). The nodes in each region are scaled relative to the known, distinct abundance of each monoamine in that region. The loops connecting these regions represent established cortico-subcortical-thalamic loops, highlighting monoaminergic regulation occurring in the brainstem, striatum, thalamus and finally the cortex feeding back in a top-down regulation of monoaminergic nuclei. The '?' represents the yet uncharacterised interregional neurochemical regulation in the brain.

gradient maintained by an ATPase pump (Justice, 1993). The acidic pH of the vesicles protects catecholamines from auto-oxidation (Mosharov et al., 2009; Umek et al., 2018). In NA neurons, the cytosolic synthesis of NA is presumably minimal and mediated by soluble DA- β -hydroxylase (DBH) (Issidorides et al., 2004; Olschowka et al., 1981). Rather, NA synthesis mainly occurs in the exocytotic vesicles, with a higher presence of soluble and membrane-bound DBH (Rush and Geffen, 1980) and DA itself. Thus, NA neurons could release DA, as demonstrated in numerous *in vivo* studies (Devoto et al., 2014, 2005; Kempadoo et al., 2016). The synthesis of 5-HT occurs from the mono-oxygenation of tryptophan into 5-hydroxy-tryptophan (5-HTP) by tryptophan hydroxylase 2 (TPH2) and its decarboxylation into 5-HT by AADC, both occurring in the cytosol (Mosienko et al., 2015). 5-HT is then transported into exocytotic vesicles via VMAT2. Monoaminergic neurons lack distinct biochemical identities, as they share the AADC and VMAT2 in their anabolic pathways.

One of the main consequences of this overlap of enzymatic/transport pathways between monoaminergic pathway occurs when precursors of these monoamines are exogenously administered. It is well documented that the exogenous administration of the precursors 5-HTP or L-DOPA - aromatic amino acids that enter the CNS and all cells via L-amino acid transporter 1 promote the synthesis and release of 5-HT from catecholaminergic and DA from 5-HTergic neurons (De Deurwaerdere et al., 2017; Jackson and Wightman, 1995; Navailles et al., 2010; Stamford et al., 1990). Consequently, the ability of 5-HTP and L-DOPA to release 5-HT and DA, respectively follows the catecholaminergic innervation for 5-HTP and serotonergic innervation for L-DOPA towards ectopic release in unexpected CNS regions (Barrière et al., 2025; Navailles et al., 2011).

Such a mechanism has been largely underestimated in the treatment of Parkinson's disease where the striatum could not be the main site of the therapeutic efficacy of L-DOPA and L-DOPA-induced dyskinesia as far as DA is involved in L-DOPA's action (De Deurwaerdere et al., 2017). Even without the injection of exogenous amino acid precursors, the blockade of key enzymes involved in the degradation of monoamines (see below) can lead to an accumulation of 5-HT in DA terminals, which can be released by striatal DA terminals upon stimulation (Zhou et al., 2005).

The degradation of biogenic amines is also complex, involving several enzymes and cell types. Among these, monoamine oxidase (MAO) plays a central role in regulating amine levels within the brain. The enzyme exists in two isoforms, MAO_A and MAO_B, which catalyse oxidative deamination and produce hydrogen peroxide, ammonia and the corresponding aldehyde. Catecholaminergic neurons mainly contain MAO_A, whereas 5-HT neurons are rich in MAO_B at their terminals. DA, NA, and 5-HT are degraded by MAO_A and to a lesser extent DA by MAO_B (Di Giovanni et al., 2016; Ramsay, 2013). Another enzyme catechol-O-methyltransferase (COMT) can also metabolize DA and NA (Eisenhofer et al., 2004). COMT is not present in the DA terminals of rodents (Kaenmaki et al., 2009; Myohanen et al., 2010; Schendzielorz et al., 2013), implying that glial cells or other neurons uptake extracellular DA in tissues where the DA clearance is low (Fig. 2). COMT is also involved in the degradation of metabolites generated by MAO. The final product from these two distinct catabolic pathways is homovanillic acid (HVA) for DA metabolism and vanillylmandelic acid (VMA) for NA metabolism (Eisenhofer et al., 2004).

In catecholaminergic neurons, DA is metabolized to DOPAL by the activity of MAO_A and to 3,4-dihydroxyphenylacetic acid (DOPAC) by

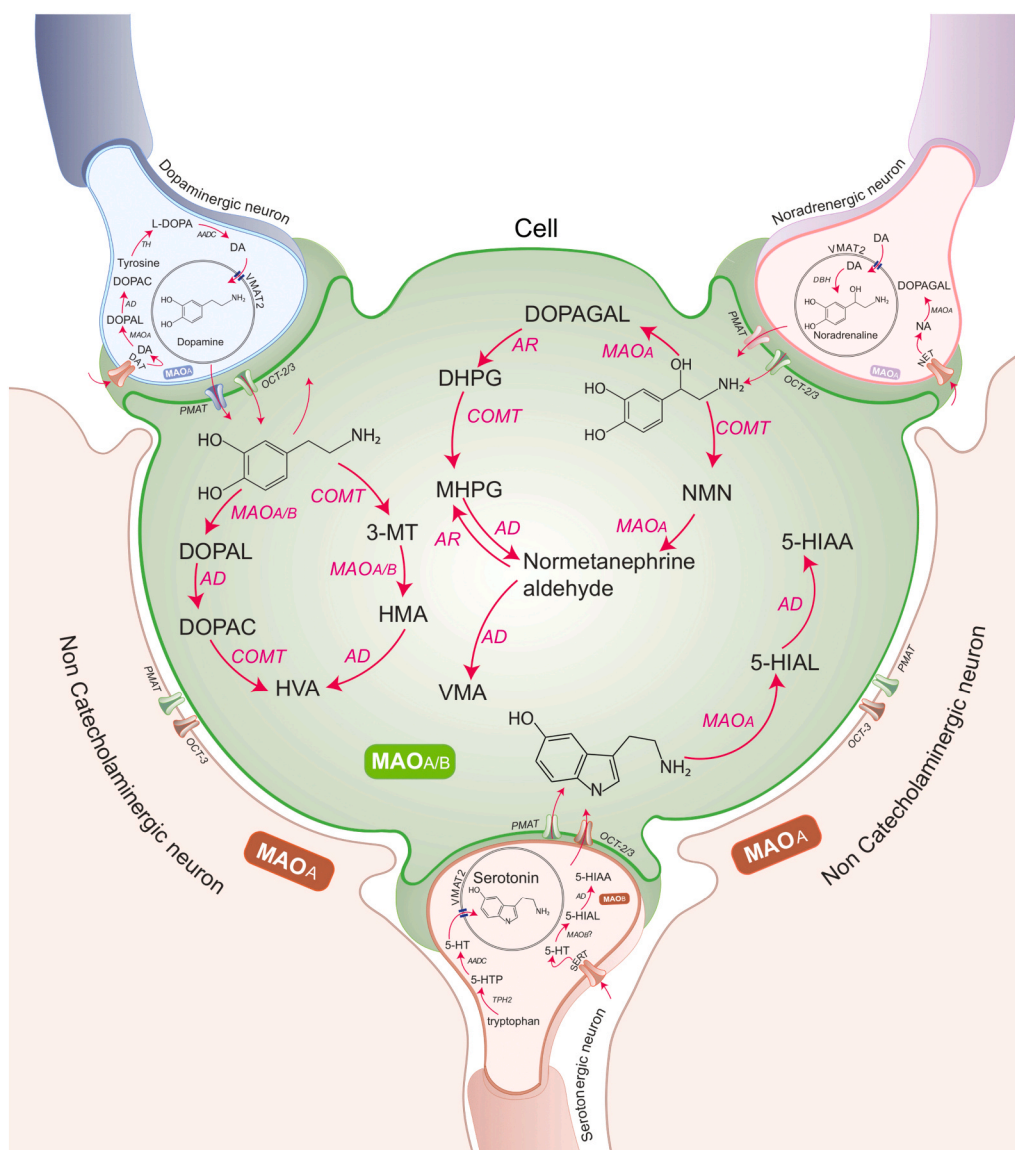


Fig. 2. Metabolic pathways of monoaminergic neurons. This drawing represents terminals of dopamine (DA) (top left), noradrenaline (NA) (top right) and serotonin (5-HT) neurons surrounding a glial cell and non-catecholaminergic neurons. L-amino acid decarboxylase (AADC); L-3,4-dihydroxyphenylalanine (L-DOPA) tyrosine hydroxylase (TH); tryptophan hydroxylase 2 (THP2); Monoamine oxidase A and B (MAOA, MAOB); 3,4-Dihydroxyphenylacetaldehyde (DOPAL); Aldehyde dehydrogenase and reductase (AD, AR); 3,4-Dihydroxyphenylacetic acid (DOPAC); Catechol-o-methyltransferase (COMT); 3-Methoxytyramine (3-MT); homovanillic acid (HVA); dopamine beta-hydroxylase (DBH); 3,4-methoxyphenylacetaldehyde, (DOPEGAL); 3,4-dihydroxyphenylglycol (DHPG); 3-Methoxy-4-hydroxyphenylglycol (MHPG); vanillylmandelic acid (VMA); 5-hydroxytryptophan (5-HTP); 5-hydroxyindole acetaldehyde (5-HIAL); 5-hydroxyindoleacetic acid (5-HIAA); vesicular monoamine transporter 2 (VMAT2); DA transporter (DAT); 5-HT transporter (SERT); NA transporter (NET); plasma membrane monoamine transporter (PMAT); organic cation transporter 2 and 3 (OCT-2, OCT-3). (VMAT2); DA transporter (DAT); 5-HT transporter (SERT); NA transporter (NET); plasma membrane monoamine transporter (PMAT); organic cation transporter 2 and 3 (OCT-2, OCT-3).

aldehyde dehydrogenase (AD) enzymes (Goldstein et al., 2015, 2013; Liu et al., 2014). It is then converted to HVA by COMT, outside of catecholaminergic neurons at least in rodents. Likewise, NA is metabolized to VMA by MAOA, AD, and COMT. The NA metabolites 3-methoxy-4-hydroxyphenyl ethylene glycol (MHPG) and 3,4-dihydroxyphenyl ethylene glycol (DHPG) are produced from the activity of MAOA followed by aldehyde reductase (AR) (Eisenhofer et al., 2004). 5-HT is metabolized to 5-hydroxyindole acetic acid (5-HIAA) by MAOA and AD.

A third layer of complexity, already evoked above, is that terminals of monoaminergic neurons, and particularly the DA ones, do not have the same biochemical organization in different CNS regions. It is also important to consider that the regulations reported here, mostly from rodents, are different in primates, vertebrates, and invertebrates (see below).

3.2. The biochemical index of monoamines: raw data and indirect indexes of turnovers

Depending on the analytical conditions, monoamines, their precursors, and their metabolites can be measured, and the ratios between them can be calculated. One may wonder what the meaning is of these indices measured using different biochemical approaches. The extracellular levels, corresponding to release from terminals, are minimal. Even though extracellular DA is relatively high in the striatum, high sensitivity is required to measure it in dialysates. The contribution of these extracellular levels to the whole measurement of DA in post-mortem striatal tissue is considered negligible (Pregeljc et al., 2019). The proportion of DA in the cytosol corresponds to approximately 3%, whereas the main contribution to the DA tissue content likely comes

from exocytic vesicles. This research again focuses on DA in striatal tissue, although it is likely different in extrastriatal territories as well as for 5-HT and NA, as their metabolism (turnover) is higher compared to DA in the striatum. Extracellular levels of DA are extremely low in the prefrontal cortex when compared to extracellular levels of DA in the striatum (factor 10) (Gobert and Millan, 1998; Gobert et al., 2000). In addition to faster cortical metabolism compared to the striatum, regional monoamine differences are related to the possible participation of NA terminals (Devoto et al., 2005) and the lower density of DA reuptake sites, including the norepinephrine transporter (NET) and the DA transporter (DAT, negligible in the cortex) (Holloway et al., 2018). The extracellular levels of NA and 5-HT are also low in numerous brain regions (Devroye et al., 2017; Gobert et al., 2000; Navailles et al., 2011; Sharp et al., 1989). The values are in the femtomole/sample range.

One advantage of postmortem analysis is to address the heterogeneity of monoamines quantity across multiple brain regions without addressing the specific biochemical regulations in the different brain regions. Table 1 reports values in ng/mg for NA, DA, and 5-HT in numerous brain regions from different rat and mouse strains (mostly males) for naïve, wild type, or animals receiving saline (*i.e.* control groups). These differences could be attributed to species, strain, age, or sex, but are also related to the precise location of the punch. The gross distribution of tissue content for monoamines indicates high levels of DA in the striatum (often > 1 ng/mg), lower levels in the mesencephalon and amygdala (approximately 0.1–0.3 ng/mg), and low levels in most cortical regions and the hippocampus. The 5-HT concentrations were higher in the VTA, SN, DRN, and MRN. The concentrations of NA were higher in the hypothalamus, thalamus, DRN, and VTA. The concentrations of NA in the hypothalamus are among the most variable parameters between studies. The noradrenergic innervation of the hypothalamus is extremely heterogeneous (Kvetňanský et al., 1983), which could explain the large variability between studies.

Monoamine metabolism also varies between brain regions, as exemplified by the DOPAC/DA [sometimes (HVA + DOPAC/DA)] and 5-HIAA/5-HT ratios (Bannon et al., 1981; Commissiong, 1985; Ebinger et al., 1987; Fitoussi et al., 2013; Johnston and Moore, 1983; Shannon et al., 1986), whose functional value for assessing metabolic activity is greater than that of the tissue neurotransmitter itself (Eisenhofer et al., 2004). This statement may not be as evident when performing correlative analyses, which is, in part, the purpose of this review. These metabolic indices are often termed turnover, a biological process that includes the balance between the synthesis and catabolism of monoamines (Eisenhofer et al., 2004; Glowinski, 1975). Turnover *stricto sensu* has been less studied *ex vivo* or *in vivo*, as it requires specific analyses that mainly rely on radiolabelled compounds. Therefore, the ratios are indirect indexes of turnover only. Again, the same index, such as DOPAC/DA, likely has a different meaning in the striatum than in extrastriatal tissue. In the striatum, the origin of DOPAC would be primarily related to DA terminals themselves in basal conditions in rodents (Justice, 1993; Sharp et al., 1986; Zetterström et al., 1988). The reason for less knowledge in extrastriatal tissue is related to the difficulty in reliably measuring low quantities of DA and its metabolites. Moreover, DA-based pharmacology (psychostimulants, DA agonists, and DA antagonists), which has enabled researchers to better understand striatal DA terminal function and activity, is less relevant in several extrastriatal tissues due to the low levels of DAT and D₂ autoreceptors (Jiao et al., 2003). This argument alone should justify broader analyses of tissue content for neurotransmitters and metabolites across the CNS, as the regulatory mechanisms operating on the activity of monoaminergic neurons in the CNS are likely to vary across brain regions.

3.3. The neuron and the gland

Varicosities along the axons of monoaminergic neurons interact with local partners, including cell bodies, glial cells, and afferent neurons/dendrites. The biochemical activity of monoaminergic terminals is

modulated by these partners (heterologous controls) and autoregulatory mechanisms (homologous controls) that maintain an activity set point for varicosities originating at the neuronal level. Biochemical modulation of terminal activity can be independent of the electrical activity of cell bodies, as exemplified by the ability of synaptosomes to synthesize and release neurotransmitters in exocytotic and non-exocytotic manners (de Belleruche and Bradford, 1980). These considerations are important when correlating the tissue content of neurotransmitters between brain regions. While homologous controls maintain activity at similar levels, heterologous controls can change according to the conditions. It is possible that changes in the correlation profiles between brain regions we will describe later correspond to subtle variations in heterologous controls rather than gross changes in homologous controls that would modify neurotransmitter or metabolite quantities across regions. It is remarkable that the full lesion of DRN/MRN 5-HT neurons does not change the striatal tissue content of DA three weeks after the lesion (De Deurwaerdere et al., 1998, 1995), despite the multitude of evidence showing that 5-HT modulates the activity of DA neurons (Alex and Pehek, 2007; De Deurwaerdere and Di Giovanni, 2017; Kapur and Remington, 1996; Soubrie et al., 1984). In fact, the electrolytic lesions (MRN, DRN, or both) modified the indirect index of the turnover of DA (DOPAC/DA) sometimes in an opposite manner between the cortex and the NAc (Herve et al., 1981; Hervé et al., 1979a). Thus, numerous homeostatic processes occur to maintain tissue DA levels, with distinct mechanisms between brain regions. The corollary would be that similar tissue levels in various conditions do not imply that the parameter has not changed qualitatively. For this reason, correlative approaches could add information beyond classical quantification, and are seen as a complementary analysis in this schema. However, it is important to address some technical issues to ensure that the biological variability is relevant because the correlative analyses are still not associated with a hypothesis-driven framework.

4. Practical and technical measurements of post-mortem data, with possible caveats

In quantitative approaches, variability can be attributed to multiple factors, including analytical devices. Therefore, analytical procedures are extremely rigorous whenever quantification is required for tissue content or *in vivo* extracellular dialysate levels. Variability in these quantitative data also arises from biological factors, some of which can be controlled (e.g., circadian rhythm), whereas others remain inherently uncontrollable (inter-individual variability).

The expression of group data as a mean or median masks the variability of individual values, whereas correlative and multivariate approaches give more importance to each individual value within the group (Butler et al., 2025a). Therefore, the number of animals in each group must be sufficient to address the scientific question. If the question is simply the validation and the extent of the ablation of serotonergic neurons, for instance, *n* = 5 may be sufficient. However, the characterisation of correlative networks between genetic or behavioural phenotypes, or pharmacological treatments warrants an *n* sufficient to meaningfully capture inter individual variability. We set a somewhat arbitrary minimum of *n* = 8 for this purpose. Although, attributing biological meaning to a correlation becomes problematic if the observed variability arises from factors such as injection errors or aberrant sampling locations.

A rigorous approach is crucial in quantitative and semi-quantitative methods because quantification is relative, not absolute. Since multivariate analysis emerges from this type of quantitative data, there are still no clear expected profiles of neurochemical connectivity. Thus, the recommendations reported in Fig. 3 are rigorously followed to avoid pitfalls that would introduce unwanted variability in a dataset at the expense of biological variability.

A first source of variability is the experimental design (Fig. 3A). It is essential to manage the various biases that could influence the dataset

Table 1

Tissue concentrations of monoamines across the rodent brain. Data compiled from the control groups of various studies are referenced in the table.

Brain region	Tissue concentration (ng/mg)			
	NA	DA	5-HT	ref
Orbital Cortex	0.12–0.24	0.01–0.06	0.16–0.22	(Bharatiya et al., 2020; Butler et al., 2025b; Gros et al., 2021; Marty et al., 2023)
medial	0.09–0.19	0.03–0.14	0.05–0.156	(Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013)
lateral	0.01–0.132	0.005–0.02	0.05–0.144	(Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013)
dorsolateral	0.07–0.182	nd–0.005	0.05–0.205	(Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013)
Frontal Cortex				
Prelimbic	0.04–0.26	0.02–0.06	0.05–0.16	(Bharatiya et al., 2020; Butler et al., 2025b; Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013; Gros et al., 2021; Marty et al., 2023)
Infralimbic	0.05–0.305	0.03–0.123	0.07–0.14	(Bharatiya et al., 2020; Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013; Gros et al., 2021)
Cingular anterior	0.06–0.58	0.016–0.08	0.05–0.23	(Bharatiya et al., 2020; Butler et al., 2025b; Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013; Gros et al., 2021; Marty et al., 2023)
Cingular posterior	0.09–0.24	0.015–0.05	0.07–0.11	(Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013)
M2	0.07–0.287	0.015–0.036	0.04–0.13	(Bharatiya et al., 2020; Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013)
M1	0.17–0.17	0.01–0.013	0.165–0.21	(Butler et al., 2025b; De Deurwaerdere et al., 2022)
Insular anterior	0.1–0.33	0.013–0.062	0.06–0.18	(Bharatiya et al., 2020; Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013)
Insular posterior	0.09–0.312	0.006–0.04	0.1–0.18	(Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013)
Cortex				
Somesthetic	0.18–0.23	0.005–0.018	0.05–0.08	(Butler et al., 2025b; De Deurwaerdere et al., 2022; Gros et al., 2021)
Auditory	0.15–0.16	0.009–0.013	0.07–0.08	(Aman et al., 2025; Butler et al., 2025b)
Visual	0.14–0.19	0.008–0.009	0.046–0.055	(Butler et al., 2025b; De Deurwaerdere et al., 2022)
Retrosplenial	0.12	0.015	0.14	(Gros et al., 2021)
Entorhinal	0.2	0.01	0.12	(A. Gros et al., 2021)
Basal Ganglia				
Nucleus accumbens	0.09–0.13	1.6–3.5	0.13–0.3	(Aman et al., 2025; Butler et al., 2025b; Gros et al., 2021)
Accumbens Shell	0.06–0.32	1.1–3.8	0.11–0.311	(Bharatiya et al., 2020; Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013)
Accumbens core	0.03–0.34	0.84–4.1	0.06–0.14	(Bharatiya et al., 2020; Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013)
Striatum	0.09	10.565	0.35	(Aman et al., 2025)
Dorsomedial	nd–0.23	2.8–10	0.006–0.2	(Bharatiya et al., 2020; Butler et al., 2025b; Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013; Gros et al., 2021; Marty et al., 2023)
Ventromedial	nd–0.2	2.6–7.5	0.11–0.26	(Bharatiya et al., 2020; Butler et al., 2025b; Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013; Gros et al., 2021; Marty et al., 2023)
Dorsolateral	nd–0.08	2.8–10	0.07–0.17	(Bharatiya et al., 2020; Butler et al., 2025b; Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013; Gros et al., 2021; Marty et al., 2023)
Ventrolateral	nd–0.22	2–14	0.07–0.26	(Bharatiya et al., 2020; Butler et al., 2025b; Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013; Gros et al., 2021; Marty et al., 2023)
Globus Pallidus	nd	0.06	0.2	(De Deurwaerdere et al., 2022)
STN	0.1–0.3	0.04–0.1	0.04–0.36	(Bharatiya et al., 2020; Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013)
EPN	0.1–0.3	0.05–0.1	0.12–0.23	Fit, Dede, Chag, Bar (Bharatiya et al., 2020; Chagraoui et al., 2019; De Deurwaerdere et al., 2020; De Deurwaerdere et al., 2022)
Hippocampus				
dorsal	0.05–0.25	nd–0.03	0.03–0.33	(Aman et al., 2025; Bharatiya et al., 2020; Butler et al., 2025b; Chagraoui et al., 2019; De Deurwaerdere et al., 2020; De Deurwaerdere et al., 2022; Fitoussi et al., 2013; Gros et al., 2021; Marty et al., 2023)
ventral	0.13–0.34	0.004–0.019	0.08–0.24	(Aman et al., 2025; Bharatiya et al., 2020; Butler et al., 2025b; Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013; Gros et al., 2021; Marty et al., 2023)
Amygdala	0.14–0.8	0.1–0.24	0.19–0.45	(Aman et al., 2025; Butler et al., 2025b; Gros et al., 2021; Marty et al., 2023)
Basolateral	0.1–0.36	0.09–0.6	0.09–0.25	(Bharatiya et al., 2020; Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013)
Central	0.12–0.25	0.06–0.185	0.09–0.25	(Bharatiya et al., 2020; Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013)
Diencephalon				
Thalamus	0.17–0.4	0.009–0.012	0.1–0.12	(Aman et al., 2025; Bharatiya et al., 2020; Gros et al., 2021; Marty et al., 2023)
mediodorsal	0.08–0.55	0.015–0.05	0.02–0.15	(Butler et al., 2025b; Gros et al., 2021)
ventrolateral	0.07	0.006	0.11	(Butler et al., 2025b)
Anterior dorsal	0.133	0.022	0.076	(De Deurwaerdere et al., 2022)
n. reticularis	0.141	0.056	0.114	(De Deurwaerdere et al., 2022)
posterior	0.375	0.082	0.28	(De Deurwaerdere et al., 2022)
ventrobasal	0.136	0.042	0.07	(De Deurwaerdere et al., 2022)
dLGN	0.18–0.2	0.005–0.034	0.134–0.2	(Butler et al., 2025b; De Deurwaerdere et al., 2022)
Habenula	0.06–0.08	0.06–0.015	0.04–0.13	(Bharatiya et al., 2020; De Deurwaerdere et al., 2020)
Hypothalamus	0.3–2	0.04–0.08	0.15–0.2	(Aman et al., 2025; Butler et al., 2025b; Gros et al., 2021)
dorsal	0.04–0.7	0.02–0.07	0.05–0.2	(Bharatiya et al., 2020; Chagraoui et al., 2019; De Deurwaerdere et al., 2020)
ventral	0.15–0.64	0.02–0.15	0.02–0.72	(Bharatiya et al., 2020; Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Marty et al., 2023)
Mesencephalon				
Sup Colliculus	0.16–0.17	0.04–0.08	0.18–0.35	(Butler et al., 2025b; De Deurwaerdere et al., 2020)
VTA	0.24–0.47	0.22–0.5	0.23–0.7	(Bharatiya et al., 2020; Butler et al., 2025b; Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013; Gros et al., 2021; Marty et al., 2023)
SN	0.1–0.26	0.2–0.4	0.32–1	(Bharatiya et al., 2020; Butler et al., 2025b; Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013; Gros et al., 2021; Marty et al., 2023)
Red nucleus	0.24	0.1	0.22	(Gros et al., 2021)
DRN	0.23–1.2	0.05–0.08	0.23–1.15	(Bharatiya et al., 2020; Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013; Gros et al., 2021)

(continued on next page)

Table 1 (continued)

Brain region	Tissue concentration (ng/mg)			ref
	NA	DA	5-HT	
MRN	0.25–0.32	0.015–0.08	0.1–1.2	(Bharatiya et al., 2020; Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Fitoussi et al., 2013; Gros et al., 2021)
Pons - Medulla				
Inferior olive	0.24	0.01	0.1	(Gros et al., 2021)
Locus coeruleus	1,8	0.13	0.18	(Gros et al., 2021)
Cerebellum	0.14–0.28	0.01–0.035	0.02–0.06	(Butler et al., 2025b; Gros et al., 2021)

The data were obtained from studies performed on rodents. This corresponds mostly to male rats, except for Aman et al., (2025) and Butler et al., (2025b), which were performed in female and male C57Bl6/J mice. Rat strains were Wistar-Han (Fitoussi et al., 2013), Long Evans (Gros et al., 2021) or Sprague-Dawley (the other studies).

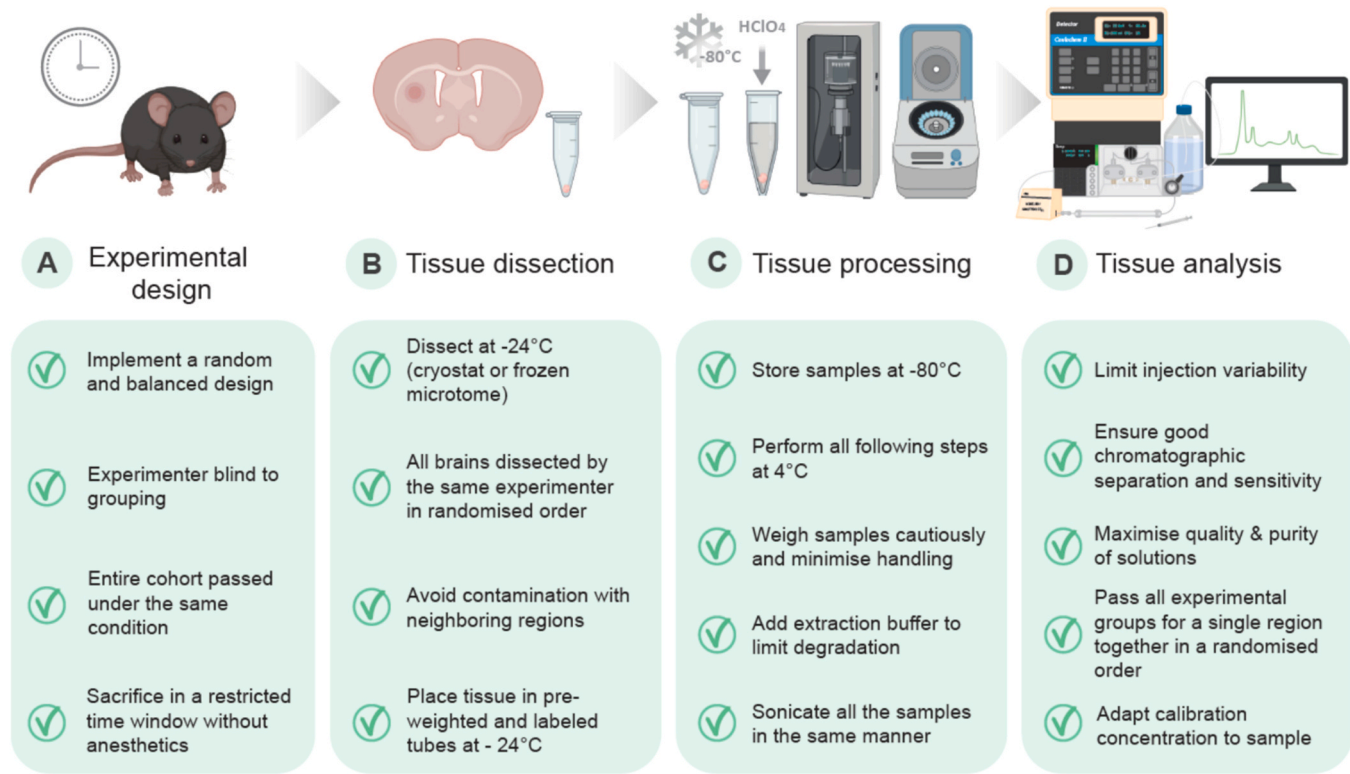


Fig. 3. – Practices to minimise non-biological variability at each experimental stage for post-mortem neurochemical quantification.

such as sex, age, strain, or life conditions are among the evident conditions that are classically managed. The time of sacrifice is another possible bias, as the concentration of monoamines can vary during the circadian cycle (Cagampang et al., 1993; Hery et al., 1977; Menon et al., 2019; Rueter and Jacobs, 1996). Thus, choosing a short time window for sacrifice limits the impact of the circadian rhythm on the dataset. In any case, a random and balanced design should be adopted. The metabolism of monoamines and classical neurotransmitters is rapid and sensitive to anesthetics (Müller et al., 2011). Thus, the sacrifice of the animals should occur without anesthesia if subtle changes in monoamine quantities are to be assessed. The experimenter should be blind to the experimental groups from animal manipulation through dissection to sample handling (injection and analysis). The number of animals should be determined before the experiment. Increasing the number of observations (n) after several weeks or months is rarely satisfactory, as the added values could differ by 10–20% compared to the initial controls. Such variability arises from all steps of the procedure (life conditions, sampling procedures, and other chromatographic/detection conditions).

A second source of variability comes from the selection of tissues. To further adhere to a rigorous scientific method, the same experimenter should dissect the CNS regions of interest (Fig. 3B). The CNS regions of

interest can be dissected directly after removal. In this case, the pieces of tissue are usually quite large (several mg), which allows for their direct weighing before freezing. The main limitation of this approach is the risk of damaging neighbouring structures and the limited number of brain regions that can be dissected. This is likely the cause of the regularly reported elevated levels of DA in the cortex (above 100 pg/mg of tissue) and sometimes in the hippocampus, likely from the presence of striatal or mesencephalic tissue in the sample. Thus, even 1% of striatal tissue can compromise the determination of cortical or hippocampal DA quantity (Table 1). Since this type of regional contamination is unexpected, it can vary between samples in the same experiment without necessarily implying outliers but inducing non-biological variability. Alternatively, the frozen whole brain can be dissected using a cryostat or frozen microtome at approximately -24°C (Fitoussi et al., 2013; Saa-vendra, 1977). This method offers several advantages: precise control over the size of the tissue 'punches', smaller tissue samples, larger number of brain regions, picture of extracted regions. In the hands of a trained manipulator, this technique limits the punching of neighbouring regions. This approach has also some limits as regards the tissue conditioning.

The third source of variability is tissue conditioning (Fig. 3 C). Tissues should be stored in labelled tubes at -80°C. The tissue samples

must then undergo a process of compound extraction (Fig. 3 C). Whenever small pieces of tissues are taken from the frozen brain in a cryostat (punches), there are two issues to consider with this method of sampling. The tissue weight, low (1–3 mg), is determined by the difference between filled and empty tubes with specific procedures (Dellu-Hagedorn et al., 2017). The second point is related to the multiple steps of tissue handling before processing. At variance with peripheral tissues, the extraction of monoamines from CNS tissues can be achieved using perchloric acid (HClO₄, 0.1 N), without absorption/filtration, after sonication. All procedures, including subsequent centrifugation, were performed at 4°C. The procedures should be homogenous and reproducible for all tissues.

The last source of variability comes from the analytical devices (Fig. 3D). The sensitivity and stability of the analytical devices are checked daily using standard solutions. Stability also implies a reproducible injection procedure, minimizing the injection volume error and dispersion of the sample before the HPLC column. A more complicated source of variability is confounding peaks in the chromatogram. The quality and purity of each constituent is important to minimize contaminations of different origins. Contaminants can also be compounds present in tissue samples. In fact, most HPLC conditions can detect several compounds that are not measured because their endogenous concentrations are low or negligible. When measuring DA, DOPAC, or HVA tissue content in regions with low endogenous levels, such as the hippocampus or cortex (Table 1), a higher gain is required (i.e., the detector amplification factor applied to the electrochemical signal). Particularly at higher gains, negligible concentrations of electrochemically sensitive compounds can induce a signal, in addition to a higher background than the baseline. Therefore, caution is required to ensure that the peak of the target compound does not coincide with that of a tissue ‘contaminant’. For instance, under our conditions, L-DOPA, 5-HTP, and adrenaline, among others, induce strong electrochemical signals in standards, but their concentrations are low or negligible (adrenaline in the brain). At a higher gain, we can measure low but reproducible quantities of L-DOPA or 5-HTP (5–50 pg/mg of tissue). Therefore, the correct separation of a sample under the initial conditions should include the determination of the elution time of other compounds in the sample, whether measured or not (Marty et al., 2023). The concentration of the pairing ion (for instance sodium octyl sulfate) or the modification of the pH of the mobile phase should allow for adjusting the elution time of the compounds to achieve a clear separation between peaks (Chagraoui et al., 2019). Once the chromatographic conditions are satisfactory, calibration curves are performed with 3–5 distinct concentrations in the expected range of tissue concentrations for each compound (Table 1). These curves enable the determination of detection limits, quantification limits, and saturation points.

The tissue quantities of compounds of interest vary between brain regions; therefore, it is preferable to study the samples by region for all experimental groups. Each brain region should be analysed using the same analytical device and from the same batch of dissected regions.

Once the chromatographic conditions are satisfactory, calibration curves are performed with 3–5 distinct concentrations in the expected range of tissue concentrations for each compound (Table 1). These curves enable the determination of detection limits, quantification limits, and saturation points.

Thus, the recommendations reported in Fig. 3 are rigorously followed to avoid pitfalls that would introduce unwanted variability in a dataset at the expense of biological variability. Since multivariate analysis emerges from this type of quantitative data, there are still no clear expected profiles of neurochemical connectivity.

5. Correlative and multivariate analyses of CNS neurotransmitters

5.1. Rational to change paradigms

Monoaminergic systems exhibit biological responses through their actions across the CNS (De Deurwaerdere and Di Giovanni, 2017; Fitoussi et al., 2013; Hensler et al., 2013). Knowledge of the biological functions and complex interactions between monoamine systems is limited and does not sufficiently consider the widespread nature of their axonal branches. It is postulated that changes in monoamine tone in one brain region can have effects on the activity of the other monoamines, both locally and distally, directly or indirectly via other neurotransmission systems (Fig. 1). This has been exemplified in numerous dynamic, *in vivo* studies without providing a wide-brain overview (De Deurwaerdere and Di Giovanni, 2017; Guiard et al., 2008; Hensler et al., 2013).

Quantitative neurochemical data from post-mortem tissue offers the possibility to approximate interaction between neurotransmitters and/or their metabolism in distinct CNS regions, sometimes in brain regions that are never considered. As indicated above, the a correlative approach provides qualitative insights into neurochemical tissue contents that can occur independently from quantitative changes and whenever quantitative changes occur, it can reveal the qualitative aspects accompanying these changes that may be obscured when expressed as mean + /- SD (Butler et al., 2025a). Brain-wide correlative approaches are being used for a variety of non-quantitative data in neuroscience. Functional magnetic resonance imaging (fMRI), magnetoencephalography (MEG), and multielectrode array (MEA) data are analysed using correlative approaches between brain regions, furthered by the application of graph theory to describe network properties (Bullmore and Sporns, 2009; Wang, 2010). These works often describe ‘functional connectivity’ that is, regions whose signals correlate over time. It has resulted in the description of several ‘brain networks’, regions whose fMRI signal tends to correlate in given states, notably the Default Mode Network (DMN), active during rest or internally directed cognition (Raichle, 2015). To conceive neurochemical networks a Pearson/Spearman correlation, according to normality testing, is performed between two sets of data from the same animals, significant ($p < 0.05$) r values are included in a correlogram usually without correction for multiple comparisons, discussed in 5.7.2. Fig. 9 shows a graphical depiction of the representation of correlations within a correlogram.

5.2. Is there a default mode neurochemical network?

We have performed several studies using a correlative approach to study tissue neurotransmitter content. The pattern of correlations between monoamines and their metabolites has mostly been studied at rest, comparing pharmacological or genetic manipulations. In our experience, there are few significant correlations across the brain in vehicle-treated/naive rodents under resting conditions (Bharatiya et al., 2020; De Deurwaerdere et al., 2020, 2020; Dellu-Hagedorn et al., 2017; Fitoussi et al., 2013; Marty et al., 2023; Puginier et al., 2019). Other groups conducting similar analyses have also reported sparse correlations across a limited number of brain regions in control animals (Midzyanovskaya et al., 2020; Mlost et al., 2024). Even in invertebrates, we found relatively few monoamines correlations between the cerebroid ganglion, pedal ganglion, and gonads in *Perna perna* taken along the Moroccan Atlantic coast (Fig. 5)(Klouche et al., 2015). Consistently, the density of correlations in resting conditions was $\leq 10\%$ for any monoamine or metabolite. The organization of these correlations is not reproducible between experiments; therefore, this approach would be devoid of quality control, except for the quantitative values. For instance, in the two experiments concerning the effects of the 5-HT_{2C} receptor agonists WAY-163,909 and lorcaserin, the experimental

conditions were similar (Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Di Giovanni et al., 2020; Whitestone et al., 2019). In both experiments, vehicle-treated rats had ~7% density of correlations, but the organization of these “basal” correlations was distinct for all neurotransmitters (see Fig. 4).

The lack of a ‘default’ set of inter-regional correlations (correlation stability across different cohorts) in resting conditions may be due to the variability in ‘resting conditions’ prior to sacrifice.

5.3. Expression of the data and quality controls

Before discussing the validity of a correlative neurochemical approach, we address the data representation. In the initial submission of the first article that utilized correlative analysis (Fitoussi et al., 2013), we drew figures reporting only significant Pearson correlations for all three monoamines, alone or in combination, between regions, as seen in a recent article (Mlost et al., 2024). Nonetheless, as raised by one reviewer at that time, these systems are chaotic in essence, and the

correlative links may not be clearly expected. Thus, full maps reporting the coefficients of correlations that are statistically significant are likely to be more representative. As discussed above, the maps are, in most cases, full of blank boxes for monoamines under resting conditions. We take this as a criterion of resting states, even though the inter-regional correlations are not reproducible.

There are two other possible control criteria, although they do not constitute a proof of concept. First, correlations between neurotransmitters and their metabolites (s) in the same region can often occur. These are never full positive correlations for all CNS regions, but the proportion of significant correlations between parent neurotransmitter and metabolite in individual region can reach up to 50% of brain regions (Dellu-Hagedorn et al., 2017; Marty et al., 2023). Second, when combining the correlations between neurotransmitters, the proportion of correlations tends to be higher within a specific brain region than across brain regions (Chagraoui et al., 2019; Fitoussi et al., 2013). The occurrence of correlations between monoamines in a single region has also been reported in the human striatum (Hortnagl et al., 2020)

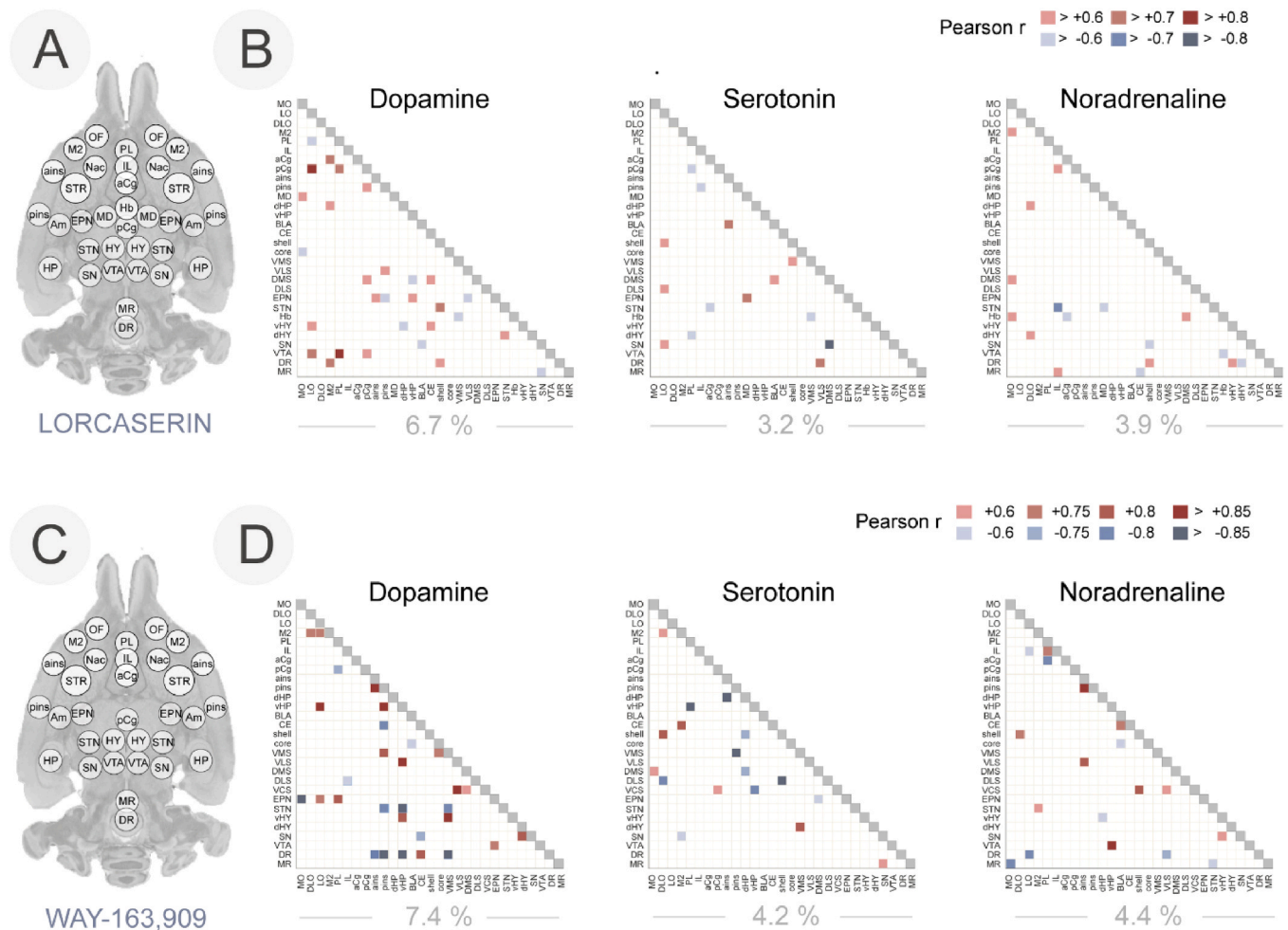


Fig. 4. – Neurochemical correlations of monoamines in resting conditioning in vehicle-treated rats from two distinct experiments relative to lorcaserin (Di Giovanni et al., 2020) and WAY-163909 (Chagraoui et al., 2019). Approximate punches localization for the 30 and 29 brain regions. (B, D) Correlative maps reporting existing, significant correlations (Pearson’s r coefficient) for the tissue content of dopamine, serotonin, and noradrenaline (pg/mg) between the different brain regions sampled in vehicle-treated rats. Each colored box corresponds to a significant correlation ($p < 0.05$; red, positive; blue, negative), and the color intensity represents the strength of the correlation (r values). The percentages of significant correlations for each system are indicated below the correlation maps. Orbito-frontal cortical regions (OF): medial orbital cortex (MO), lateral orbital cortex (LO), Dorsolateral orbital cortex (DLO); Motor cortex M2; Prelimbic cortex: PL; Infralimbic cortex (IL); Anterior cingulate cortex (aCg); Posterior cingulate cortex (pCg); Anterior insular cortex (ains), Posterior insular cortex (pins); Nucleus accumbens (Nac): shell and core; Striatal regions (STR): Dorsomedial striatum (DMS), Dorsolateral striatum (DLS), Ventrolateral striatum (VLS), Ventrocaudal striatum (VCS); Amygdala (Am); Basolateral nucleus (BLA), Central nucleus (CE); Habenula (Hb); Mediodorsal thalamic nucleus (MD); Entopeduncular nucleus (EPN); Subthalamic nucleus (STN); Hypothalamus (HY): dorsal parts (dHY), ventral parts (vHY); Hippocampus (HP): dorsal anterior parts (dHP), ventral posterior parts (vHP); Substantia nigra (SN); Ventral tegmental area (VTA); Dorsal raphe nucleus (DR); Median raphe nucleus (MR).

hippocampus, or dorsolateral prefrontal cortex (Di Maio et al., 2026), an interesting finding that could anticipate possible correlations between extracellular levels whenever multiple neurotransmitters are measured. A recent study has shown that in the cortex of the rhesus macaque, extracellular levels of neurotransmitters monitored by microdialysis and tandem mass spectrometry correlate differently according to cognitive demand (Mayer et al., 2025). These two points are not strict criteria, as a correlative link may emerge depending on the physiological status of the animals.

The distinct neurotransmitter correlations in macaque monkeys, depending on cognitive status, is important information to suggest that the links between neurotransmitters can vary as a function of brain state and organism-environment interactions. The correlation between parent compounds and their metabolites is also modulated by environmental and genetic factors, as previously reported (Aman et al., 2025; Gros et al., 2021; Marty et al., 2023). In the human prefrontal cortex, there was no quantitative difference in post-mortem quantities of monoamines in patients with schizophrenia compared to age-matched controls; however, there were qualitative changes, as revealed by a correlative analysis (Di Maio et al., 2026).

5.4. Connection or connectivity?

Beyond compounds relationships within a region the benefit of a brain-wide approach would be to assess the relationship between neurochemicals in distinct regions, where significant correlations could be considered a form of connectivity. Similar to the ‘functional connectivity’ described in fMRI datasets when the signal in two regions correlates temporally in a single subject, we could conceive a ‘neurochemical connectivity’ where within a cohort a neurochemical or pair of neurochemicals correlate between regions.

The neurochemical connectivity would be distinct from anatomical connection. It would be tempting to speculate that the physical link between the SN and the striatum would favour the occurrence of correlations for DA and its metabolites between the two brain regions (originating from the same group of DA neurons). Such correlations are not reported. In fact, it has been shown for a long time that the regulatory mechanisms operating at the level of the striatum and the SN on DA content and metabolites, marked by a distinct ratio DOPAC/DA, are different despite corresponding to the same group of neurons (Argiolas et al., 1982). Thus, the correlative link of biochemical parameters

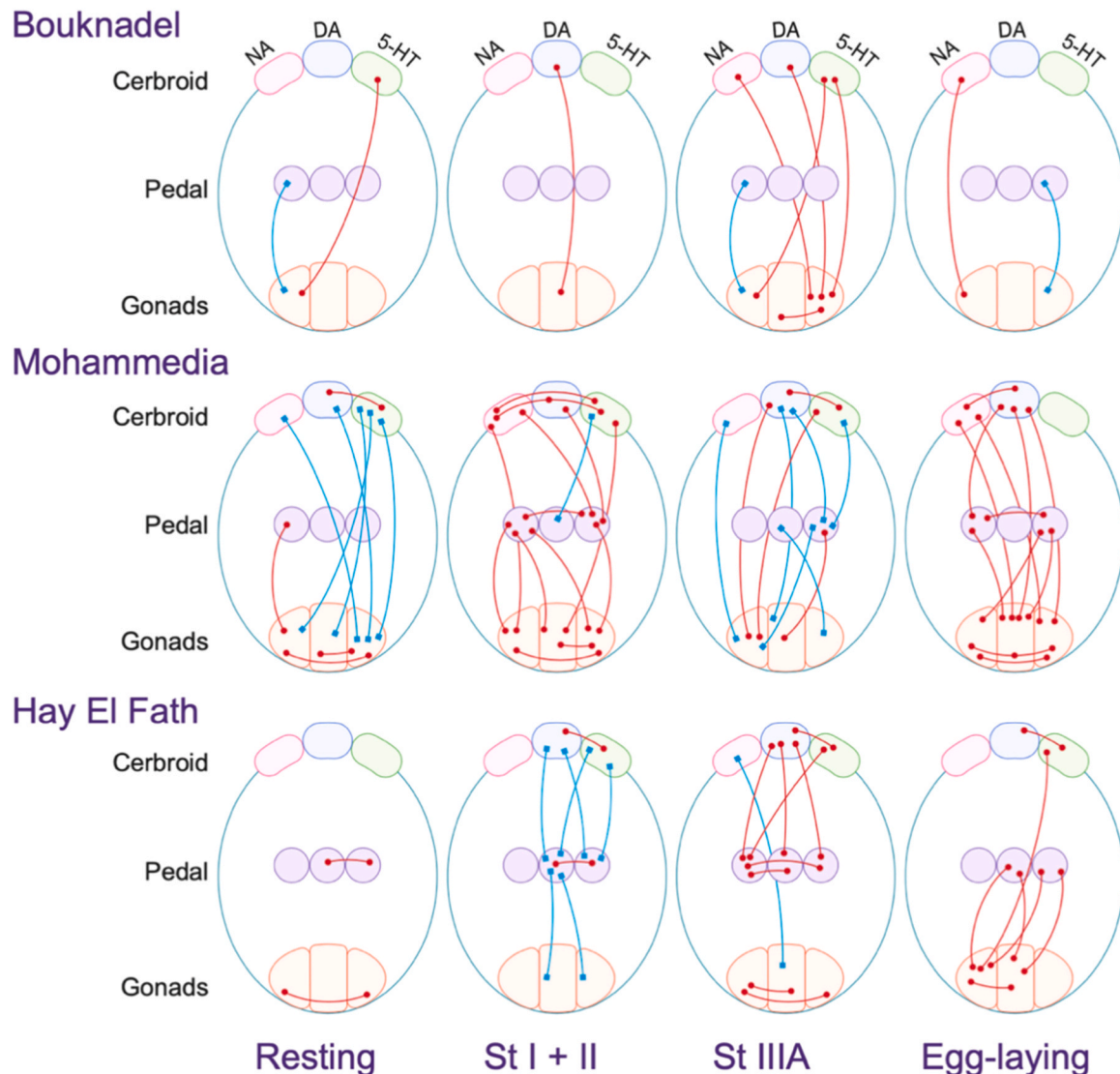


Fig. 5. Tissue monoamine connectivity in *Perna perna*. Correlation analysis of NA (left), DA (middle), and 5-HT (right) content between organs at different stages of the sexual cycle of *Perna perna* at the Moroccan sites of Bouknadel, Mohammedia, and Hay El Fath. The different stages (Resting; St I + II; St IIIA and egg-laying) are approximate. The Spearman's rank order correlation test was used to establish correlations ($p < 0.05$) between individual measures. These correlations were either positive (red) or negative (blue); adapted from (Klouche et al., 2015).

between the regions housing cell bodies and their terminal fields is not a valid criterion, additionally exemplified by the usual lack of correlations between 5-HT in the DRN and the numerous brain regions it innervates (Figs. 4–6). Of note, as acknowledged in subsection 2.2 (serotonergic neurons), measured 5-HT and its metabolite can come from at least two distinct nuclei in several brain regions, which could challenge the expectation that 5-HT content at terminal regions could correlate with 5-HT content in the DRN or MRN. More recently, we also found a lack of specific correlations for NA between punches encompassing the LC and other parts of the brain (Gros et al., 2021). A refinement of this approach would be to study the left and right hemispheres separately, as in most cases, lateralized regions have been combined. This combination could limit the occurrence of correlations if lateralized regulation occurs, as suggested by studies separating the left and right sides (Butler et al., 2025b; Mlost et al., 2024).

Therefore, correlating neurotransmitter content across the brain could correspond to the characterization of connectivity within a neurotransmitter system or between neurotransmitter systems, rather than anatomical connections. As an example of functional neurochemical connectivity between brain regions, we could evoke the presumably opposite role of cortical DA on subcortical DA function (Deutch, 1993; Latagliata et al., 2023; Louilot et al., 1989; Tassin et al., 1978). We previously reported negative correlations between the infralimbic cortex and the shell of the nucleus accumbens in a cohort of 35 Wistar rats (Fitoussi et al., 2013), but this negative correlation was not systematically found in other experiments. We could also address the ascending loop pathways linking the shell of the nucleus accumbens to the dorsolateral striatum (Haber et al., 2000), which has no functional counterpart. We reported a few correlations of the DOPAC/DA ratio between striatal quadrants in some datasets (Dellu-Hagedorn et al., 2017). This organization was lost in young R6/1 mice, a mouse model of Huntington's disease when compared to wildtype littermates (Puginier et al., 2019). Yet, the correlative links for DOPAC/DA between striatal quadrants are not systematic between experiments.

The question of what is behind an identified correlation in terms of biological meaning remains presently elusive. The study of tissue monoamines in *Perna perna* throughout different stages of the reproductive cycle was very interesting on that matter (Fig. 5). Indeed, the organization of the pattern of correlations between NA, 5-HT, and DA levels in the cerebroid ganglia, pedal ganglia, and gonads has been studied throughout the reproductive cycle. The data corresponded to the

description of what was supposedly occurring for monoamines, from the control of gonad NA function at rest to the control of gonad 5-HT function for the release of gametes (Klouché et al., 2015). The meaning of some negative correlations could also be coherent with the endocrine controls of NA or 5-HT content in the gonads (Fig. 5). The dramatic changes in the reproductive cycle and the profile of correlations at polluted sites could be attributed to an alteration in monoamine function, together with the inclusion of mussels at different reproductive stages, as the distribution of mussels at different reproductive stages was less clear at polluted sites (Klouché et al., 2015).

Thus, more than physical connections, correlative analysis would inform on neurochemical connectivity/coherence, a connectivity at rest that is still elusive for neurotransmitters, and potentially more informative for the ratios DOPAC/DA and 5-HIAA/5-HT in rodents. It is still based on the data obtained from individuals composing the cohort.

5.5. Lack of clear correlation profile: inter-individual differences?

Correlative approaches, and multivariate analyses rely on individual values. Apart from the variability coming from non-biological sources, as discussed above, individual differences in a cohort likely play a role in the lack of a systematic profile of correlations at rest. Although this has not been specifically addressed, our database indicates that the composition of the cohort is a factor conditioning the pattern of correlations. In a cohort of 35 Wistar male rats, animals were characterized by their behavioral performances in six tests addressing impulsive/compulsive dimensions of behavior (Dellu-Hagedorn et al., 2018). For each test, three groups were composed a posteriori: performant, intermediate, and non-performant. The intermediate group was discarded and the monoamine profiles between performant and non-performant rats were compared, i.e. the extreme phenotypes within the cohort. The composition of the performant and non-performant groups was distinct for each of the 6 tests, indirectly showing that a maladaptive response in one test does not imply a maladaptive response in all tests (Dellu-Hagedorn et al., 2018). We reported changes in tissue 5-HIAA/5-HT, 5-HT, DOPAC/DA, and DA between performant and non-performant rats for each test, sacrificed one month after the last behavioural test. We are still processing these data, but the preliminary analysis indicates that the pattern of the correlations for the ratios is completely distinct between the two cohorts in the “risk taking” test (Butler et al., 2025a). Acknowledging that all the data from these

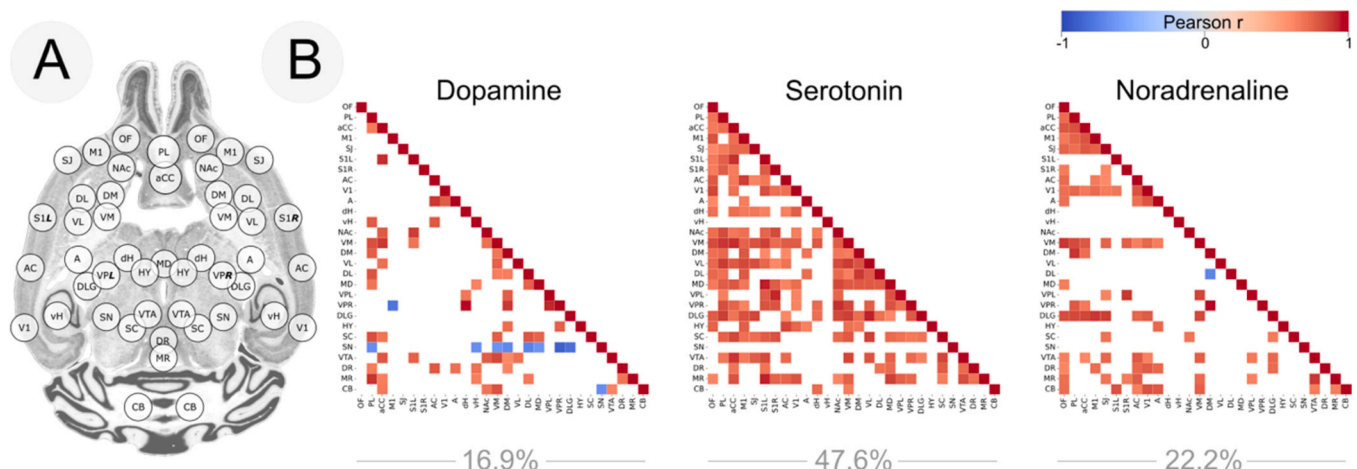


Fig. 6. Neurochemical connectivity of monoamines following exploratory behaviour in vehicle-treated mice, data adapted from previous work (Butler et al., 2025b). OF (orbitofrontal cortex), PL (prelimbic cortex), aCC (anterior cingulate cortex), M1 (primary motor cortex), SJ (somatosensory cortex jaw), S1L/R (left/right primary somatosensory cortex), AC (auditory cortex), V1 (primary visual cortex), A (amygdala), dH/vH (dorsal/ventral hippocampus), NAc (nucleus accumbens), DM/DL (dorsomedial/dorsolateral striatum), VM/VL (ventromedial/ventrolateral striatum), MD (medial dorsal nucleus), VPL/R (left/right ventral posteromedial nucleus), DLG (lateral geniculate nucleus), Y (lateral hypothalamus), SC (superior colliculus), SN (substantia nigra), VTA (ventral tegmental area), DR (dorsal raphe nucleus), MR (median raphe nucleus), CB (cerebellum). The percentages below correlation matrices report the density of correlations over the entire matrix.

animals were initially combined to study correlations for monoamines (Fitoussi et al., 2013) or their metabolism (Dellu-Hagedorn et al., 2017), this suggests that inter-individual differences play an important role in the outcome of correlation profiles. Consequently, under resting conditions, a specific profile cannot be expected.

Because there is no specific profile of correlations characteristic of vehicle-treated animals, correlation profiles obtained between two or three cohorts, such as transgenic mice versus wild type or drug versus vehicle, require cautious interpretation. In two independent experiments, the 5-HT_{2C}R agonists lorcaserin and WAY-163909 reduced the number of correlations for DA tissue content and increased the number of correlations for NA and 5-HT (Chagraoui et al., 2019; De Deurwaerdere et al., 2020; Di Giovanni et al., 2020). In contrast, the number of correlations for the DOPAC/DA ratio was dramatically reduced by lorcaserin and slightly increased by WAY-163,909 (De Deurwaerdere et al., 2020; Whitestone et al., 2019). Both 5-HT_{2C}R agonists had a few but distinct quantitative effects, except for an increase in tissue DA in the central amygdala.

5.6. Resting conditions or activated conditions?

It has been shown for years that mice subjected to open field or electric foot shock stress exhibit an increase in the cortical ratio DOPAC/DA (more DOPAC, less DA) immediately after exposure (Hervé et al., 1979b; Tassin et al., 1980). It has been shown that it corresponds to an increase in extracellular levels of DA and NA (Abercrombie et al., 1989; Feenstra, 2000; Gresch et al., 1994). Regardless of the existence of this robust effect in rodents, one may wonder what the consequences of this effect are in other parts of the brain. This line of questioning is one proposed purpose of correlative analysis. Instead of considering exclusively resting conditions, it would be interesting to study correlative maps of neurotransmitters under conditions of behavioural demands, here referred to as “activated” conditions.

We have only one example in which a transient exposure to a new environment was used to address the mechanism of action of one psychedelic drug, the new environment favouring unconditioned exploratory behaviour (Halberstadt and Geyer, 2018). Such a situation was interesting for us to determine the extent to which the correlative maps of neurotransmitters are sensitive to the environment of the animal,

sacrificed immediately following forced exposure to new environment. We studied the effect of the 5-HT_{2A}R agonist TCB-2 and the role of 5-HT_{2A}R in its effect using the 5-HT_{2A}R antagonist MDL-100,907. After having monitored the head twitches to determine if the treatments were efficacious, we changed the flooring for the 10 last minutes of the experimentation. We substituted a classical Perspex floor for a novel flooring made up of different textures and colours to promote ‘forced’ exploratory behaviour in mice. We were surprised to discover that the number of correlations, mostly positive, in vehicle-treated mice was tremendously high for 5-HT content across the brain, with a density of almost 50% (Fig. 6). This was also true for DA and NA which additionally showed a clear regional organization. We also found extremely high level of positive correlations for amino acids including GABA, aspartate, and glutamate with the highest density of correlations for glutamate (>60%) (unpublished data). When combining pairs of neurotransmitters across the brain, the profile obtained was organized whenever DA or NA were included, suggesting that these two systems play an important role in organising neurotransmitter responses to the novel environment.

TCB-2 dose-dependently reduced the number of correlations for all neurotransmitters and suppressed the organized pattern of correlations whenever DA and NA were included. MDL-100,907 alone also dramatically reduced the number of correlations. With TCB-2 (high doses) or MDL-100,907, there was an increased occurrence of negative correlations (see Fig. 7A). In contrast to TCB-2, MDL-100,907 did not induce major quantitative effects on the tissue concentrations of neurotransmitters, as expected from previous studies (De Deurwaerdere et al., 2021; De Deurwaerdere and Di Giovanni, 2017). The dramatic decrease in neurochemical connectivity within and between neurotransmitter systems with 5-HT_{2A}R ligands suggests that 5-HT_{2A}R plays a role in the organization of neurotransmitter systems in response to a novel environment. However, the quantitative analyses still indicated that TCB-2 did not selectively act on 5-HT_{2A}R because some of its effects were not altered by MDL-100907. Thus, other receptors are likely involved in the reduction of correlations induced by TCB-2 (Butler et al., 2025b).

In another, unpublished experiment, we studied the effect of the typical antipsychotic haloperidol on monoamine tissue content. However, the density of correlations was extremely low in vehicle-treated

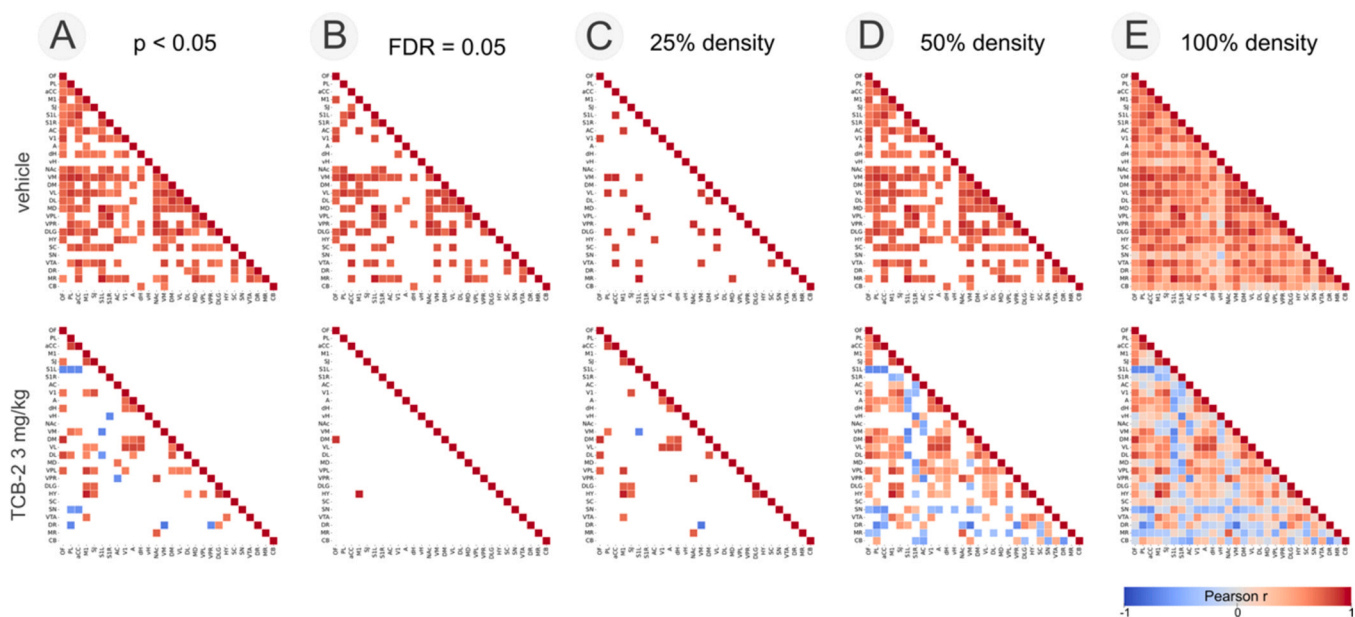


Fig. 7. – Variable methods of correlation selection. All correlograms show inter-regional 5-HT networks in vehicle (top) or 3 mg/kg TCB-2 (bottom) treated animals. A shows all significant correlations ($p < 0.05$), B significant correlations after Benjamini-Hochberg correction for multiple comparisons with a false discovery rate (FDR) of 0.05. Columns C, D and E with density thresholds of 10%, 50% and 100% applied. Pearson r is indicated by the colour bar.

animals, masking any loss of organized correlations. Therefore, it seems pertinent to address the effect of potential antipsychotic drugs (haloperidol, 5-HT_{2C}R agonists) and probably other classes of medicines in more complex experimental settings rather than ‘resting’ states.

5.7. Extension of connectivity: graph theory and neurochemical networks

To better investigate and eventually characterize the neurochemical connectivity after forced exploratory behaviour and 5-HT_{2A}R ligands, we expanded our correlative analysis into graph theory. The application of graph theory to correlative approaches in neuroscience is not new and is well-suited to conceptualizing the relationships between brain regions (Bullmore and Sporns, 2009).

The output of the correlation analysis is a matrix of *r* values (Pearson or Spearman), which can be filtered for significance *i.e.* $p < 0.05$. Most commonly, this is represented as a heatmap (correlogram) and compared visually (Figs. 4 and 6) or by making a heatmap of the difference between two correlograms. Differential correlograms are most common in the field of neuroimaging, highlighting altered brain regions (Carhart-Harris et al., 2016, 2012), although they are limited to the comparison between two conditions, often pre- and post-condition within a subject. Rather than a direct comparison between the two states, the characteristics of one state may be of particular interest. From the application of graph theory we have seen the characterisation of specific regions or subsets of regions whose fMRI signal correlates in particular experimental contexts, *i.e.* the DMN (Raichle, 2015). Graph theory allows the construction of networks from inter-regional correlative analysis to display, analyse and characterise functional ‘brain-networks’ (in contrast to anatomical) in different experimental contexts, across different data types.

5.7.1. Graph theory and networks for post-mortem neurochemical data

A network or graph, is a mathematical structure used to model relations between objects (nodes). In the case of post-mortem neurochemistry, we are modelling the relation between any two brain regions’ tissue content of a compound or pair of compounds. Each brain region sampled is a node in the network, which is connected to other nodes via edges, correlations. Additional information can also be represented in different types of networks. Weighted networks have a value ascribed to each edge, often the *r* value of a correlation. In the case of neurochemical networks, correlations between pairs of neurotransmitters may be represented in directed multi-graphs. The use of directed multi-graphs is to maintain information in the absence of reciprocal correlations. For instance, DA in region 1 may correlate with 5-HT region 2, but the reverse is not true. Therefore there would be a directed edge from the node of region 1 to region 2 but not the reverse (Fig. 9B). Once created these networks have various graph-theoretical metrics that can be compared, the simplest of which is network density, the percentage of edges present out of all possible edges, previously referred to as the density of significant correlations. Additionally, node degree, or node strength for weighted graphs, assigns a value to each node relative to its degree of connectivity, therefore nodes (regions) with more correlations have a higher node degree. This allows the identification of hubs, highly correlated or ‘connected’ brain regions, and peripheral regions, regions whose neurotransmitter systems do not correlate with others, acting in isolation. There are more complex topological network parameters, such as clustering coefficient, small-worldness, or global and local efficiency that can be extracted; however, a key challenge will be determining which graph-theoretical metrics can be meaningfully applied to correlative networks, given that many graph-theoretical metrics were not originally designed for correlative networks.

Network analysis of quantitative regional neurochemistry is a novel field, and the physiological underpinnings of these neurochemical networks are yet to be understood. As neurochemical network analysis continues to develop, a key challenge will be determining which graph-theoretical metrics can be meaningfully extracted. Given the exploratory

nature of this neurochemical network topology metrics, the primary objective is to characterise how basic network properties (density, node degree, and cautiously global efficiency) are altered by different experimental conditions in efforts to understand their significance at the physiological and behavioural levels.

These network metrics provide a quantitative method to compare networks across treatments, moving beyond a visual inspection of correlograms. This is particularly useful in neurochemistry, where multiple networks can be generated from relatively few compounds, allowing analysis both within and between the neurochemical systems.

5.7.2. Filtering relevant correlations: handling multiple comparisons

The main caveat with this approach is the selection of correlations to include. Within the same neurochemical experiment each treatment group should have the same brain regions, and therefore the same nodes. However, this is not necessarily true for the number of significant correlations, which can be altered pharmacologically as shown in Fig. 7 A (Butler et al., 2025b). This article compared networks of significant ($p < 0.05$) correlations, without a correction for multiple correlations, revealing a dose dependant reduction in correlations (density) with TCB–2. When Benjamini-Hochberg correction for multiple comparisons was applied (FDR=0.05) there were almost no remaining correlations outside of the vehicle-treated mice (Fig. 7B), and for DA and NA no treatment with a density $< 2\%$. As the general loss of strong positive correlations remains evident when looking at all correlations (100% density; Fig. 7E), supporting the disorganisation of neurochemical networks regardless of significance handling, we understood that an FDR correction may remove biologically relevant correlations (type–2 errors) and is likely not the best handling of this data.

5.7.3. Density normalisation and meaningful network parameters

To characterize the visually apparent regional organization of correlations between neurotransmitters, we assessed the variability of the node degree. Visually organized systems (vehicles) exhibited a greater standard deviation of the node degree and strength when significant correlations were observed (Fig. 8A). This means that some regions had a high number of correlations (hubs), while others exhibited little or none. However, the SD of the node degree is biased by network density, warranting cautious interpretation when density varies between treatments. This is evident when we look at density threshold networks where this decrease is not consistently significant (Fig. 8A 100% density).

Indeed, many network or graph theory metrics are influenced by network density; thus, to directly compare metrics, a normalization of network density is required. The concern here is the dismissal of biologically relevant correlations, as filtering on *p* or *r* values may not be an apt measure of biological significance. The most widely used normalization is density-matched thresholding, which sets a fixed, arbitrary density or ‘cost (*i.e.*, 20%) and includes edges corresponding to the top 20% of *r* values (Bullmore and Bassett, 2011). To avoid biases, it is best to assess parameters across a range of density thresholds, focusing on findings that persist across several densities.

The two network parameters shown in Fig. 8 exemplify this point. The left panel ‘ $p < 0.05$ ’ will be biased by variable network density while in the right ‘100% density’ each network has equal density. Looking at these two network parameters, only the global efficiency remains significant when we normalize treatment network density (at 25%, 50%, and 100% density). Global efficiency, defined by the average inverse shortest path length, is a measure of network integration often used in functional connectivity networks. In our neurochemical networks global efficiency was significantly reduced by TCB–2 across all density thresholds, as shown in Fig. 7. Therefore, the neurochemical network changes induced by TCB–2 would be best described as a loss of significant correlations (network density when using $p < 0.05$) and a reduction in global efficiency or any network parameter altered by treatment across a range of density thresholds.

5.7.4. Idiosyncratic features of neurochemical networks

Networks derived from post-mortem neurochemical data are unique compared to the conventional applications of graph theory in neuroscience, such as fMRI and other imaging techniques.

First, it is quantitative, so edges reflect first-order biological meaning. Starting with a ng/mg neurochemical topology facilitates: inter-species analysis, meta-analysis, and a new perspective to study biochemical organisation across the brain. Secondly, we are able to collect several parameters (neurochemicals) in any given region, in contrast to neuroimaging, where network analysis is applied to a single pretreated signal across the brain. This facilitated the analysis of several inter- and intra-compound networks, allowing statistical comparison of several networks per treatment, as exemplified in Fig. 8.

Additionally, we can begin to explore the relationships between parameters, *i.e.* inter-monoamine correlations, such as 5-HT – DA interactions across the brain. To address the inter-compound analysis, we implemented directed networks, as depicted in Fig. 9B. These networks have been included in Fig. 8; however, their handling, that is, the calculation of network parameters (*i.e.*, global efficiency), is modified because of the possibility of multiple edges between nodes. The analysis of the interaction between systems in this way is relatively new and warrants development. The inter-neurotransmitter networks form the TCB–2 project revealed a clear organisation of correlations when 5-HT, glutamate, GABA, aspartate and glycine were paired with either catecholamine DA or NA (Butler et al., 2025b) (unpublished data). This organisation may suggest a topological hierarchy of neurotransmitter systems in a specific experimental context, something previously relatively unexplored. This dataset reports a dose-dependent increase in the percentage of negative edges (correlations) between regions, while the vehicles of this cohort (in a forced exploratory paradigm) had sparse and highly specific negative correlations. This raises the question of the

meaning or biological underpinning of negative correlations. This negative correlation is a difficult question to address beyond the conception that the parameters in the two regions either act in concert with each other, resulting in a positive correlation, or in opposition, resulting in a negative correlation. We previously found that a specific correlation for a parameter between two regions was positive in controls and negative after treatment or conditions and vice versa (Gros et al., 2021). It was also puzzling to see in the control group of the TCB–2 experiment (thus only exposed to a new environment) that VTA DA could positively correlate with DA, 5-HT, and NA in several territories, whereas DA in the SN negatively correlated with DA, 5-HT, and NA in various regions (Butler et al., 2025b; Fig. 6). It would suggest that the exploratory behaviour favours motivation and cognition at the expense of procedural memory. This would be functional connectivity, but it is purely speculative. Correlations do not represent causality, and these curious situations recall that the interpretation of one correlation is still elusive.

Finally, we could aggregate networks into a multilayer network of neurotransmitters within an experiment, intra- or inter-compound, to identify recurrent regions and/or networks of interest. This would allow us to address how different neurotransmitter systems are organised and affected by treatment via indirect and direct correlations, the latter using partial correlations / inverse covariance (precision) matrix. This becomes particularly interesting with the development and availability of a neurochemical database upon which meta-analysis may be performed (Butler et al., 2025c). An indirect analysis has already been performed on all sampled regions. However, network parameters may also be calculated for subsets of regions to investigate more specific pathways such as somatosensory or basal ganglia functions. This would be akin to frequently described alterations in the DMN in fMRI datasets, *i.e.* a defined subset of regions of interest whose connectivity is

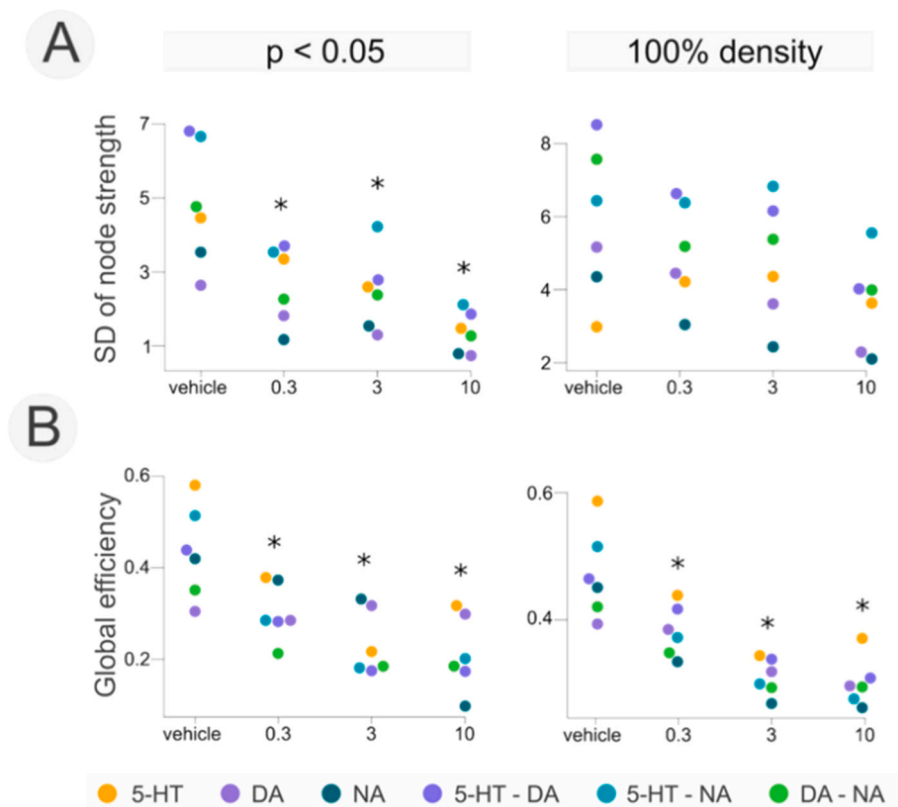


Fig. 8. – Correlation selection influences network metrics. Two network parameters were compared between treatments using either a significance threshold of $p < 0.05$ (left) or 100% density threshold (right), see Fig. 6 A and E, although was done for all conditions. The standard deviation of node strength and weighted global efficiency are shown in panels A and B, respectively, for networks within and between monoamines across the whole brain. Significance is indicated by * $p < 0.05$, Tukey's post-hoc test, following a significant one-way ANOVA.

consistently altered across conditions or states Fig. 9.

As the application of graph theory in neuroscience continues to expand, we are likely to see the refinement of approaches tailored to different data types (Bullmore and Sporns, 2009; Fornito and Bullmore, 2015). However, linking network parameters to physiological or phenomenological processes remains a challenge in post-mortem neurochemistry. As whole-brain graph analysis evolves, standardized practices will inevitably emerge; however, it will be important to remain open-minded and adaptive, particularly when applying these methods across diverse modalities.

6. Conclusion

In this article, we addressed the possibility of using postmortem quantification of neurotransmitters and their metabolites to study the brain-wide connectivity of neurotransmitter systems. These studies are sensitive to the variability of the individual values composing a cohort, and we have cited several sources of technical variability that could impact correlations without biological significance. This approach offers a unique characterization of the structure and relationship of neurotransmitter systems under various conditions.

At rest, neurochemical characterization may reveal the organization of specific behavioural traits. To study the mechanism of action of drugs, it is valuable to determine altered neurotransmitter connectivity, especially in the absence of quantitative modifications. Our recent work indicates that specific environmental conditions may be appropriate for further studying the mechanism of action of drugs, as a mobilized neurochemical system may be more sensitive to pharmacological manipulation. Furthermore, this suggests that the environment before death can have consequences on the connectivity of neurotransmitters, which may allow for the characterization of neurochemical systems mobilized for different tasks or environmental demands in the brain. The limitation of this approach is that there are no specific correlation profiles under resting conditions, acknowledging that the number of correlations is usually low. As far as biological explanations are concerned and the lack of reproducibility, we posit that the inter-individual differences among animals composing a group largely influence the outcome of correlative links, particularly at rest.

Given that analytical precision is applied at all the steps discussed above, postmortem neurochemical analysis results in a dataset with meaningful variability, permitting the conceptualization of neurochemical connectivity and the construction of neurochemical networks (Hamed and Kurs, 2018; 2020). As has been the case with ‘functional connectivity’ and functional brain networks established in fMRI research, we posit that this analysis offers a valuable lens for assessing brain-wide neurochemical states under various conditions. Correlated network analysis is exploratory in neurochemistry and requires further replication and validation. However, we believe that it could provide a complementary exploratory tool over classical quantitative analysis. With development a hypothesis generating framework, it could highlight regions whose neurochemical natures are linked under certain conditions. These descriptive approaches still need to be improved to potentially include other neurochemicals like amino acids (Hamed and Kurs, 2018; 2020; Parrot et al., 2025; Pifi et al., 2026) and implement brain wide neurotransmitters/metabolism organization.

CRedit authorship contribution statement

Chloé Aman: Writing – original draft, Visualization, Validation, Methodology, Data curation. **Jasmine J Butler:** Writing – review & editing, Writing – original draft, Visualization, Data curation, Conceptualization. **Rahul Bharatiya:** Writing – review & editing. **Philippe De Deurwaerdere:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Data curation, Conceptualization. **Abdeslam Chagraoui:** Writing – review & editing, Data curation. **Adeline Cathala:** Writing – review & editing.

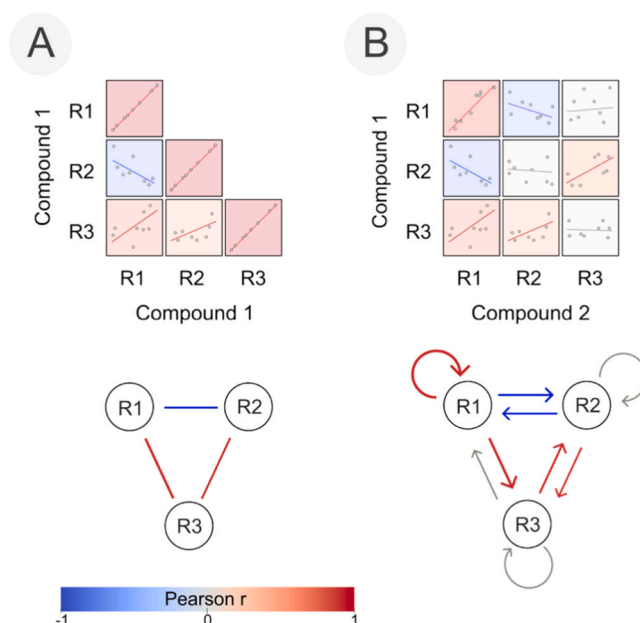


Fig. 9. – Network construction from neurochemical post-mortem data. For the three brain regions R1–3, panel A depicts an intra-compound matrix/correlation matrix of r values (top) and the corresponding undirected weighted network (bottom). B shows the equivalence between compounds 1 and 2, resulting in a directed weighted multigraph (network).

Declaration of Competing Interest

All the authors declare that they have no established conflicting financial interests or personal relationships that may have influenced the research presented in this paper.

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Data availability

No data was used for the research described in the article.

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